Credit Cards and Retail Firms: Historical Evidence from the U.S.

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Overview

This study examines the small firm not as a *borrower*, but as a *lender*—a role that small businesses frequently played in the U.S. before the rise of bank credit cards, and one they continue to play worldwide

Why it matters for policy:

- Merchants pay 2–4% interchange fees on every card transaction
- Regulators worldwide debating whether to cap these fees (EU, UK, US Credit Card Competition Act)
- But, bank cards replaced even more expensive in-house credit operations

Main finding: Bank credit cards allowed small firms to *outsource* the credit supply function, *reducing* their costs net of interchange fees

Roadmap

1. **Historical Context**: How store credit worked and why banks replaced it
2. **Aggregate Trends**: Banks displaced stores as credit suppliers since 1958
3. **Causal Evidence**: 1978 Supreme Court shock to bank card supply
    - Store lending fell, firm profits and entry increased
4. **Mechanisms**: Cost reduction (labor, risk) dominates markup compression
5. **Policy Implications**: Interchange fees vs. in-house credit costs

Historical Context

"Most Americans had a dozen or more different forms of credit. . . an account at the neighborhood pharmacy, the neighborhood grocery store, and a half-dozen other local stores"

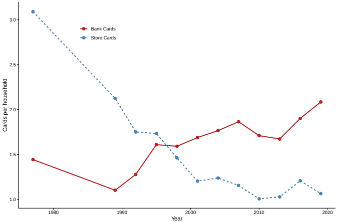
"It was the smaller merchants who first came around. . . His accounts receivable were dragging him under. . . I looked at the size of his accounts: \$4.58. \$12.82. He was sending out monthly bills."

"The bank would guarantee him payment—within days instead of months—and would take over the role of collecting from the customers. . . [while] taking a 6 percent cut."

—Bank of America salesman, 1958 (Nocera 2013)

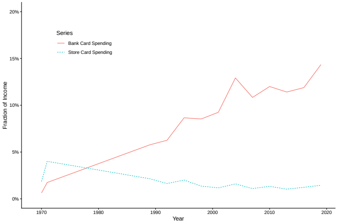
Bank Cards Displaced Store Credit

Cards per household



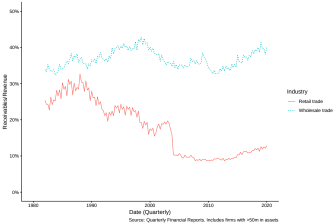
Source: SCF

Household spending



Source: SCF

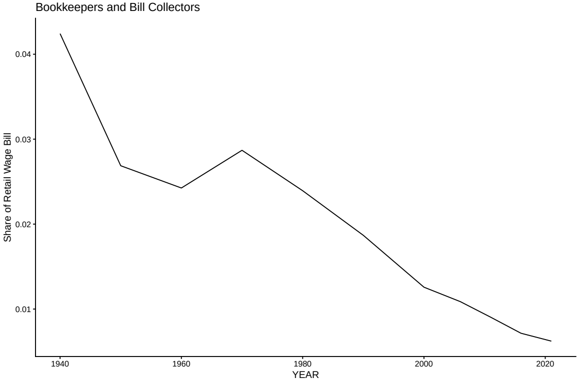
Firm receivables



Source: US Census QFR (assets $\geq$ \$50M)

Firms Saved on Credit-Related Labor

Retail firms once spent 4% of their wage bill on bookkeepers and bill collectors



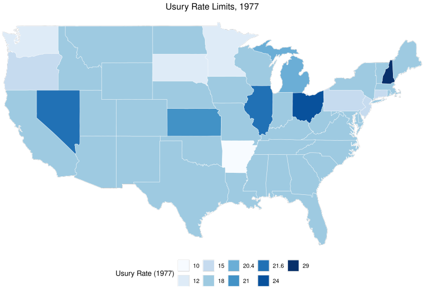
Source: US Census.

Identification: 1978 *Marquette* Decision

- Supreme Court ruled banks can lend across state lines at *home state's* usury rate
- States with usury $< 18\%$ in 1977 saw sudden increase in bank card supply from out-of-state banks
- 9 "treated" states (37M people, 16% of nation)
- Revolving credit regulated separately—no effect on mortgages, business loans
- I do *not* use the subsequent endogenous deregulation decisions: only the unexpected Supreme Court shock for identification

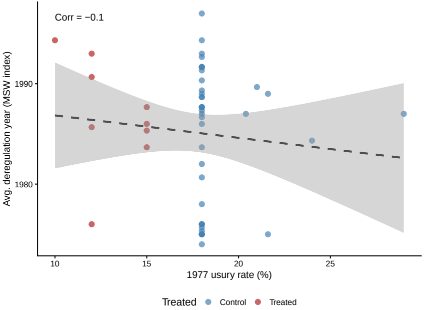
Support for Identification

No geographic pattern



Source: Cost of Personal Borrowing, 1977

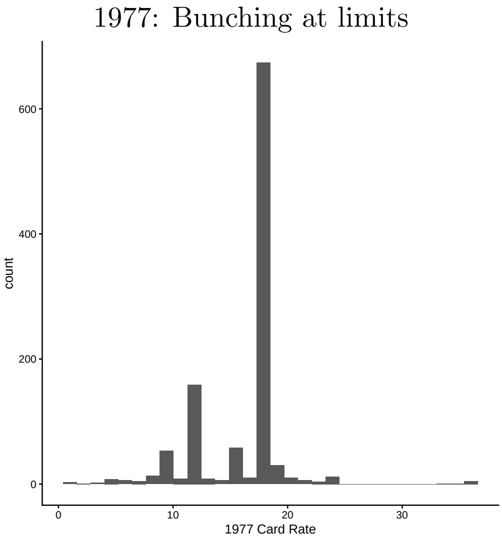
Orthogonal to other deregulation

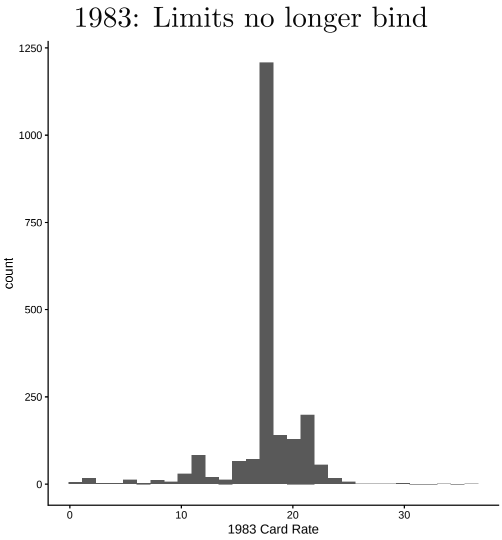


Corr = -0.10 with Mian-Sufi-Verner deregulation index

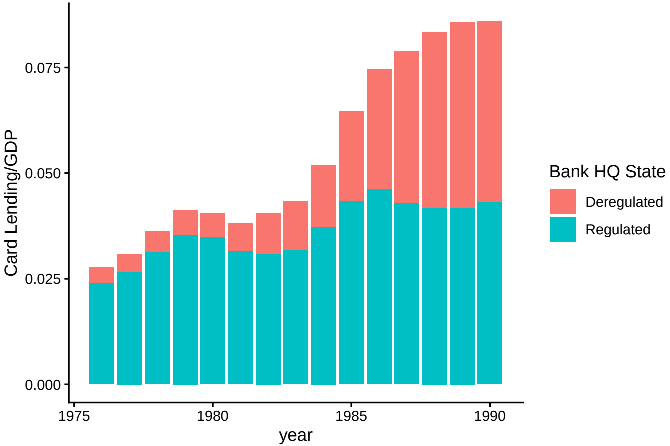
Treatment balance: no significant differences in wages, employment, GDP, or retail share between treated and control states in 1977

Usury Limits Were Binding



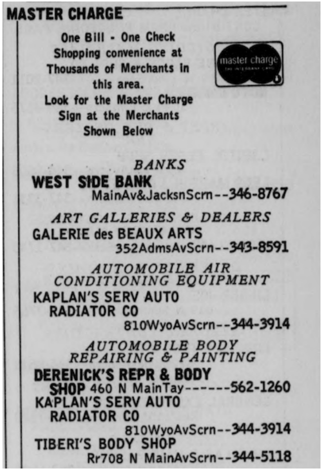


Card Lending Grew from Deregulated-State Banks



Card lending / GDP by bank HQ state usury regime. Credit card lending tripled 1976–1990; almost all growth from banks headquartered in deregulated states.

New Data on Bank Card Acceptance



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352AdmsAvScrn--343-8591

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KAPLAN'S SERV AUTO RADIATOR CO
810WyoAvScrn--344-3914

AUTOMOBILE BODY REPAIRING & PAINTING

DERENICK'S REPR & BODY SHOP 460 N MainTay------562-1260
KAPLAN'S SERV AUTO RADIATOR CO
810WyoAvScrn--344-3914
TIBERI'S BODY SHOP
Rr708 N MainAvScrn--344-5118

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Oakmont--828-4406
Verona Auto Parts Co
827AlleghenyAvOakmont--828-8500

PENN HILLS

Associated Hardware
EastHillsShopgCtr--731-4845
Penn Hills Auto Parts
119PennoakDrPitsbrgh--242-8626
Penn Hills Electronic Service
1924UniversalRdPennHills--793-3005
Russ Plumbing
11752FrankstownRd--242-0111

STANTON HEIGHTS

Associated Hardware
StantonHtsShopgCtr--661-2654

- U.S. Telephone Directory Collection, Library of Congress
- 1.97M firm-years screened in Dun & Bradstreet
- 141K covered by phonebooks across 418 municipalities
- 510 card-accepting merchant-years manually transcribed and matched to D&B
- Final sample: Compustat firms matched to D&B sales locations, with sales from D&B and card acceptance from phonebooks (2,956 firms, 283 markets, 1970–1977)

Store Lending Fell, Profits Rose

Receivables/Revenue fell

179 retail firms in Compustat

Dependent Variable:	Net Income / Revenue
Model:	(1)
Variables	
Treated × Post	0.0091*
	(0.0054)
Fixed-effects	
Year FE	Yes
Firm FE	Yes
Fit statistics	
Observations	2,000
R^2	0.21338
Within R^2	0.00058

Clustered (Firm FE) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

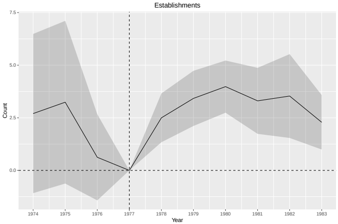
DV: net income / revenue (post-1978 level shift)

At 5%, I reject that profits fell by more than 15bp

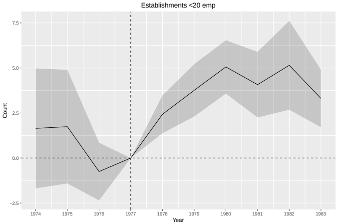
Net Entry: Triple-Difference by Industry

- **Why net entry?** Accounting profits are a flawed proxy for economic profits and are observed only for the largest firms (Compustat); net entry captures the competitive response of the whole market.
- **Why triple-difference?** Entrepreneurs also use credit cards to *fund* their businesses. The triple-difference isolates the *payment* channel: clothing retail had the highest card payment intensity, independent of card use by owners as borrowers.

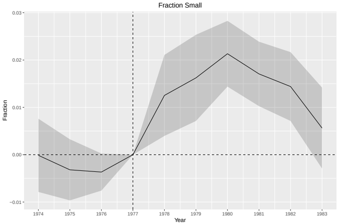
Compare to other retail sub-industries:



(a) Establishments



(b) Small establishments



(c) Fraction small

4% increase in establishments; entrants are disproportionately small.

Which Firms Adopt Bank Cards?

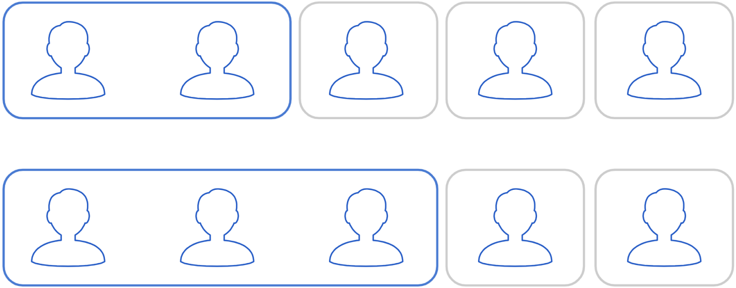
Dependent Variable:	Card acceptance	
Model:	(1)	(2)
Variables		
log(Sales)	0.7112***	0.7243***
	(0.1500)	(0.1505)
Single-establishment	0.6874***	0.7132***
	(0.2410)	(0.2308)
Firm age		-0.0075
		(0.0071)
Fixed-effects		
Industry × City FE	Yes	Yes
Fit statistics		
Observations	1,940	1,940
Squared Correlation	0.23179	0.23384
Pseudo R^2	0.24409	0.24501
BIC	1,522.3	1,528.6

Clustered (industry) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

- Geographically concentrated, high-sales firms adopt first
- Consistent with cost reduction (undiversified firms benefit most)
- Inconsistent with demand-steering (low-share firms benefit most from stealing business)

The Model in One Picture

Each face is a **retailer**; all retailers compete for customers, but *harder* against rivals that accept the **same payment method** — the grouped outlines below.



- Boxes = card-acceptance **nests**; stores inside a nest are closer substitutes.
- **Marquette** reshapes the nesting structure: as bank cards spread, more stores join the accepting nest (top row → bottom row).
- Calibration asks: **how much does the nest matter for where consumers shop?**

Calibrating to the *Marquette* Experiment

Nested logit: $u_{ij} = \delta_j + \varsigma_{g(j)} + (1-\sigma)\varepsilon_{ij}$, where $\delta_j = \alpha\, p_j + \xi_j$

- σ = fraction of taste dispersion correlated *within* card-acceptance groups
- $\sigma \to 1$: customers choose only on payment method, not what the store sells
- $\sigma = 0$: payment method irrelevant; choice is on price and idiosyncratic taste

How I calibrate: I *simulate Marquette inside the model* — raising consumer bank-card usage and merchant acceptance to match the data — and pick (σ, ζ) so the model reproduces the **actual diff-in-diff effects of Marquette**, estimated across my 252 retail markets:

1. the average sales response of treated firms ($\mathit{Treated} \times \mathit{Post}$), and
2. the *extra* sales response of firms already accepting bank cards ($\mathit{Treated} \times \mathit{Post} \times \mathit{Accept}$).

α (price sensitivity) is calibrated from the literature (and doesn't affect the answer in sensitivity analyses); only σ and ζ (the cost change) are matched. Two moments, two parameters — exactly identified.

Calibrated Model: Cost Reduction vs. Competition

Question: Does accepting bank cards mainly (a) reduce costs or (b) steal customers?

Calibrated Parameters

Parameter	Value
α (price sensitivity)	-4
σ (card differentiation)	0.017 (0.025)
ζ (cost change)	-0.105 (0.055)
Implied cost reduction	3.9%

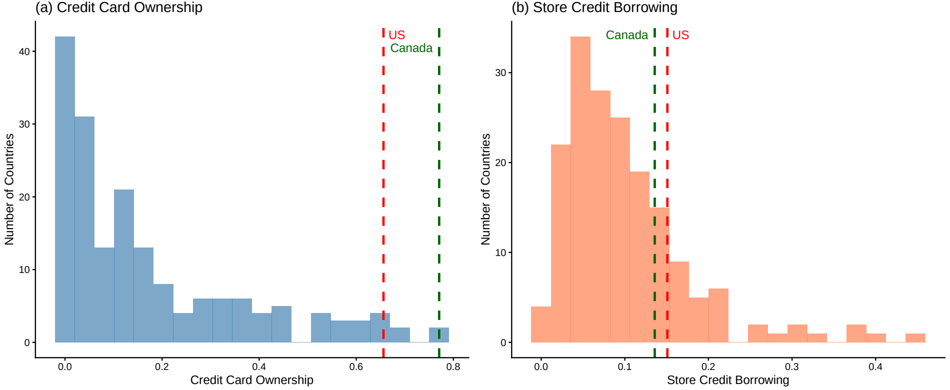
Delta-method s.e. in parentheses

Decomposition

	Cost	Markup
Baseline $\rightarrow$ Marquette	-80bp	-5bp
Cost only	-80bp	+1bp
Competition only	-16bp	-6bp

First decomposition of retail demand into product taste vs. payment-method taste: 95% CI bounds σ above at ≈ 0.07.

International Context



Many developing countries still rely heavily on store credit. The U.S. historical experience suggests that enabling bank-provided consumer credit can reduce costs for small merchants.

Conclusion and Policy Implications

Summary of findings:

- Bank cards displaced store credit, reducing merchants' labor and risk costs
- Net effect on retail industry: +4% entry, +2% profits, better recession survival
- Cost reduction ($\sim$4%) dominates markup compression (5bp)

Implications for interchange fee regulation:

- Interchange fees replaced even higher in-house credit costs
- Small merchants benefit most from outsourcing credit to banks
- Capping fees may push merchants back toward costly in-house credit
- Historical evidence: the *net* effect of card networks on merchants was positive

Backup: Sensitivity to α

α	σ	ζ	Cost reduction
-2.000	0.0157	-0.2569	-0.0951
-3.000	0.0167	-0.1499	-0.0555
-4.000	0.0173	-0.1048	-0.0388
-5.000	0.0176	-0.0802	-0.0297
-6.000	0.0109	-0.0531	-0.0197
-0.258	0.0099	-2.3915	-0.8849

Cost reduction ranges from about 2% ($\alpha = -6$) to 9.5% ($\alpha = -2$). σ is stable across specifications (0.01–0.02).

Backup: Card Acceptance by Industry

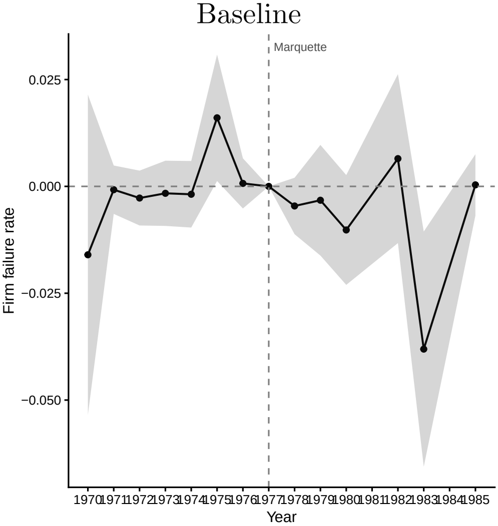
industry	acceptance	n	firms_accepting
Clothing	1.5%	4513	67
Furnishing	0.7%	4686	34
Misc.	0.6%	11982	68
Auto	0.5%	10154	46
Department Stores	0.3%	2332	7
Hardware	0.3%	3112	9
Restaurants	0.1%	12794	17
Food	0.0%	8247	2

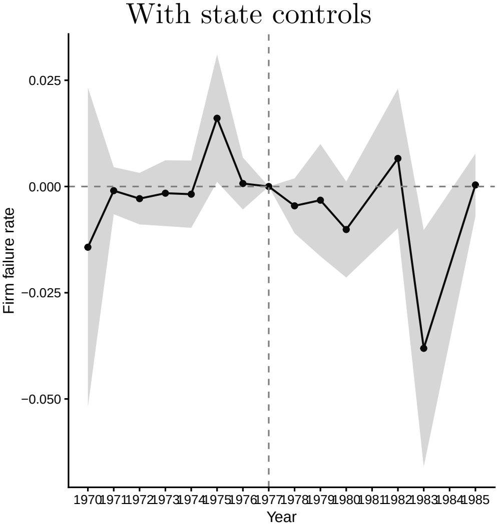
Backup: Recession Survival

Dependent Variable:	Exit rate			
Model:	(1)	(2)	(3)	(4)
Variables				
Mean log(Sales)	-0.0092***	-0.0127***	-0.0107***	-0.0155***
	(0.0004)	(0.0008)	(0.0005)	(0.0008)
Treated × Single-estab.	-0.0006	-0.0016	-0.0020	-0.0041**
	(0.0027)	(0.0026)	(0.0015)	(0.0019)
Single-estab. × Recession	0.0459***	0.0514***	0.0305***	0.0159***
	(0.0039)	(0.0046)	(0.0033)	(0.0022)
Treated × Recession × Single-estab.	-0.0159*	-0.0204**	0.0003	0.0072
	(0.0081)	(0.0099)	(0.0113)	(0.0084)
Fixed-effects				
State × Industry × Year FE	Yes	Yes	Yes	Yes
Fit statistics				
Observations	73,036	27,252	73,036	27,252
R^2	0.99726	0.99720	0.99718	0.99704
Within R^2	0.11503	0.17114	0.08924	0.12386

Clustered (STATE) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Backup: DiD with Controls





Results robust to controlling for baseline state wages and employment interacted with year FE.